Symmetry Properties of the Fourier Transform

Example 2.23 Determining an Inverse Fourier Transform Using Tables 2 and 3

Suppose that

$$X\left(e^{jw}\right)=\frac{1}{\left(1-ae^{-jw}\right)\left(1-be^{-jw}\right)}$$

Direct substitution of $X(e^{j\omega})$ into Eq. (8.130) leads to an integral that is difficult to evaluate by ordinary real integration techniques. However, using the technique of partial fraction expansion, we can expand $X(e^{j\omega})$ into the form

$$X\left(e^{jw}\right)=\frac{a/(a-b)}{\left(1-ae^{-jw}\right)}-\frac{b/(a-b)}{\left(1-be^{-jw}\right)}$$

From Theorem 1 of Table 2.2 and transform pair 4 of Table 2.3, it follows that

$$x[n]=\left(\frac{a}{a-b}\right)a^n u[n]-\left(\frac{b}{a-b}\right)b^n u[n]$$

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Example 2.24 Determining the Impulse Response from the Frequency Response

The frequency response of a highpass filter with linear phase is

$$H_{hp}\left(e^{jw}\right)=\begin{cases}0, & |w|<w_c\\ e^{-jwn_d}, & w_c<|w|<\pi\end{cases}$$

where a period of 2π is understood. This frequency response can be expressed as

$$H_{lp}\left(e^{jw}\right)=\begin{cases}1, & |w|<w_c\\ 0, & w_c<|w|<\pi\end{cases}$$

where $H_{lp}(e^{j\omega})$ is periodic with period 2π and

$$H_{hp}\left(e^{jw}\right)=e^{-jwn_d}\left(1-H_{lp}\left(e^{jw}\right)\right)=e^{-jwn_d}-e^{-jwn_d}H_{lp}\left(e^{jw}\right)$$

Using the result of Example 2.18 to obtain the inverse transform of $H_{lp}(e^{j\omega})$, together with properties 1 and 2 of Table 2.2, we have

$$h_{hp}[n]=\delta[n-n_d]-h_{lp}[n-n_d]=\delta[n-n_d]-\frac{\sin w_c(n-n_d)}{\pi(n-n_d)}$$

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Example 2.25 Determining the Impulse Response for a Difference Equation

In this example, we determine the impulse response for a stable LTI system for which the input $x[n]$ and output $y[n]$ satisfy the linear constant-coefficient difference equation

$$y[n]-\frac{1}{2}y[n-1]=x[n]-\frac{1}{4}x[n-1]$$

Impulse response $\quad x[n]=\delta[n]$

The z-transform is more useful than the Fourier transform for dealing with difference equations. However, this example offers a hint of the utility of transform methods in the analysis of linear systems. To find the impulse response, we set $x[n] = \delta[n]$; with $h[n]$ denoting the impulse response, Eq. (2.175) becomes

$$h[n]-\frac{1}{2}h[n-1]=\delta[n]-\frac{1}{4}\delta[n-1]$$

Applying the Fourier transform to both sides of Eq. (2.176) and using properties 1 and 2 of Table 2.2, we obtain

$$H\left(e^{jw}\right)-\frac{1}{2}e^{-jw}H\left(e^{jw}\right)=1-\frac{1}{4}e^{-jw}$$

or

$$H\left(e^{jw}\right)=\frac{1-\frac{1}{4}e^{-jw}}{1-\frac{1}{2}e^{-jw}}=\frac{1}{1-\frac{1}{2}e^{-jw}}-\frac{\frac{1}{4}e^{-jw}}{1-\frac{1}{2}e^{-jw}}$$

To obtain $h[n]$, we want to determine the inverse Fourier transform of $H(e^{j\omega})$. Toward this end, we rewrite Eq. (2.178) as

$$H\left(e^{jw}\right)=\frac{1}{1-\frac{1}{2}e^{-jw}}-\frac{\frac{1}{4}e^{-jw}}{1-\frac{1}{2}e^{-jw}}$$

From transform 4 of Table 2.3,

$$\left(\frac{1}{2}\right)^n u[n] \xleftrightarrow{\mathcal{F}} \frac{1}{1-\frac{1}{2}e^{-j\omega}}$$

Combining this transform with property 2 of Table 2.2, we obtain

$$-\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)^{n-1} u[n-1] \xleftrightarrow{\mathcal{F}} -\frac{\frac{1}{4}e^{-j\omega}}{1-\frac{1}{2}e^{-j\omega}}$$

Based on property 1 of Table 2.2, then,

$$h[n]=\left(\frac{1}{2}\right)^n u[n]-\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)^{n-1} u[n-1]$$